Heart Disease Prediction & Consultation System By AI/ML

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Abstract-

Heart disease is one of the leading causes of death worldwide, although it can often be treated successfully if detected at an early stage but in initial stage symptoms are very common so no one take them seriously and after a time being, these symptoms get detected when situation get irrecoverable. So, we are trying to create a model which help to detect the risk of heart disease by taking information about a person's lifestyle and other basic info like age, gender, bp, chest pain etc. However, the model only estimates the risk of heart disease. For accurate and reliable diagnosis, patients should undergo proper medical checkups and continue consulting qualified doctors.

We use the dataset which is publicly available to do some basic cleaning and preprocessing, then applied models on that dataset, after that we compared all the results based on different parameters like accuracy, recall, precision, F1 score and so on. And we found that KNN is best for detection of heart disease risk.

Index Terms **-** heart disease risk, patient health data, machine-learning models, early screening, model comparison, medical prediction, data-based assessment.

I. INTRODUCTION

Heart disease has become very common these days and most of the patients get diagnosed when their health condition become severe and treating the disease becomes even more difficult at that stage. On the other hand, initial symptoms of heart-diseases are very common which makes it very difficult to detect that disease by basic body checkups.

In recent years, many ML models are used by researchers to find patterns in the medical data of and improve the prediction of heart disease risk in patients.

We are creating a web application which help doctors and patients to detect the risk at an early stage, for which we have read existing research papers and try to implement their future scope but also we have limited resources and we have tried to keep our website easy to use.

Our aim is not to replace the doctors but to assist them as helper who can study patterns in patient records and help them to know about high-risk individuals by which they can suggest medications and lab tests accordingly. Since the initial symptoms of heart disease are often similar to those of minor illnesses, examining every patient through detailed medical tests becomes both time-consuming and expensive. As a result, it becomes difficult for doctors to identify patients who may actually be at risk of heart disease. This is where our machine learning model can assist healthcare professionals in early risk detection.

II. LITERATURE REVIEW

In last few years many researchers had worked on creating a model having very high recall value. The basic idea is that to compare the parameters and find the difference between the parameter value of a healthy person and high risk patient. Based on that difference pattern they try to build a model which have high recall value because it's comparitively ok to say a healthy person as a high risk patient but it is a dangerous thing to call any high risk patient as healthy. The parameters used by them are gender, Age, Cholesterol, Blood Sugar (Fasting), Maximum Heart Rate, ECG result. The most common dataset used by researchers again and again is UCI heart-disease dataset because it contains data of many hospitals in that most commonly used part is data of Cleveland clinical hospital, which have record of 303 patient and have very few missing values, part [14].

Most of the review paper use supervised learning method to train their model, which is like Logistic Regression, Naive Bayes, SVM, KNN, Decision Trees, Random Forests and some ensemble models. The common sentiment is that we can't say any model as perfect model, performance depends highly on the feature selection and data cleaning [20].

Some researchers say Ensemble techniques and Random forest's done quite well in handling complex and non-linear pattern. Other shows Naive Bayes and Logistic regression is liked by many doctors because it is very easy to understand it's internal functioning which is not possible in ensemble technique. Other show KNN and SVM did quite well but limitation is to it need very high tuning and data preprocessing [20].

Most of the researchers don't want to just rely on one method they combine many models to get better accuracy and prevent

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overfitting. Sometime they combine more than one dataset to oversample the dataset, in this process they found ensemble models gave better results. Also, they try to remove those features which are not affecting the result so much because with so many features model become very complex (like patient id) [21].

Some researchers don't run the experiment by themselves but use already existing heart-disease prediction model and find that most of the models train with a dataset having very small size and model accuracy highly depend on quality of data, how well it had been cleaned and which features is considered to train model. Due to small datasets, there is a high future scope to train model in huge dataset which contain varied features [14].

Most of the researchers emphasizes too much on preprocessing of data and there standard scaling. They found most of the time if scaling gets change it make so much effect on accuracy of result and performance of models. Feature selection is one of the important tasks which is as important as ML model selection because insignificant features can affect so much on Model performance because it increases the complexity of model which slow down the output [14].

Many researchers found that Model give better results on the dataset which had been used to train or test the dataset but didn't work well in real time data which is large and provided by actual hospitals, that researcher found that it is good to use Cross-Validation by which model get trained for every combination of data [20].

Researchers suggest that ML model sometime give equivalents result as of statistical models and doctors want to go with statistical model instead of ML model because statistical models are easy to understand. There is a huge future scope to create such model which is easy to understand and have high accuracy even on real time / large datasets [20].

Table 1: A comparative overview of major techniques, contributions, and challenges in Heart Disease Prediction using AI/ML.

Category	Focus	Key Contributions	Challenges
A. Traditional ML Classification Models	Predicting heart disease using classical ML (SVM, RF, DT, etc.)	Evaluates multiple ML models (Random Forest, SVM, KNN, Naïve Bayes, Decision Trees) for heart disease prediction using clinical features such as age, blood pressure, cholesterol, achieving up to ~92.9% accuracy with Random Forest [12].	Handling imbalance in medical datasets and selecting optimal features for best accuracy.
B. Feature-Selection and Model Optimization	Improving model performance via feature selection & preprocessing.	Used feature selection and compared algorithms (RF, SVM, XGBoost, MLP) on the UCI Cleveland dataset; highest classification accuracy ~94.6% with feature-selected MLP [14].	Risk of overfitting with complex models and balancing robust feature engineering with model generalizability.
C. Early Detection Using Enhanced ML Techniques	Use of advanced feature selection and hybrid strategies.	Demonstrated accurate early prediction by combining feature selection strategies with ML methods, aiming to help clinical decision making [15].	Integrating heterogeneous clinical data and ensuring model accuracy across diverse populations.
D. Ensemble & Hypertuned Model Approaches	Ensemble learning & tuning to boost predictive performance.	Applied clustering + multiple classifiers (RF, XGBoost, MLP) with hyperparameter tuning (GridSearchCV) to improve classification [16].	Computational cost of tuning models and Need for large, balanced datasets for robust performance.
E. Deep Learning on ECG Signals	Automated detection of heart diseases and arrhythmias from raw ECG signals.	Developed a deep convolutional neural network trained on large-scale ECG data to classify multiple cardiac arrhythmias with performance comparable to cardiologists. [17]	Requires very large annotated ECG datasets and Performance drops when ECG devices differ.
F. Wearable-based Heart Disease Detection	Early detection of atrial fibrillation using smartwatch sensor data.	Implemented an irregular pulse detection algorithm using smartwatch PPG data for large-scale atrial fibrillation screening, validated with ECG patches. [18]	Low prevalence leads to false positives and Confirmation with clinical ECG still required.
G. Heart Failure Readmission Prediction	Predicting 30-day unplanned readmission risk for acute heart failure patients using explainable ML.	Developed an XGBoost-based predictive model using clinical features selected via cross-validation; achieved the best AUC (~0.763), sensitivity (~0.66), and accuracy (~0.709) among tested models and used SHAP values to explain feature importance. [19]	Moderate performance (AUC ~0.763) indicates room for improvement, Limited to retrospective data and potential missing clinicalvariables.

III. METHODOLOGY

1. Data Collection and Understanding

The very first step of our project is to find a good quality dataset which don't contain so much useless features or many null values. If we choose dataset with so much null value than it will become very problematic for us to proceed to further steps as it will become very problematic to handle null values and unsignificant features. Such features may not significantly improve model performance but still consume additional computational resources. Therefore, while selecting a suitable heart disease dataset, we first identified the important features required for prediction. We selected a dataset containing relevant medical attributes such as:

Patient Id, Age, Gender, Chest Pain Type, Cholesterol, Blood Sugar (Fasting), Maximum Heart Rate, ECG result, Blood Pressure, Thalassemia, Number of Major Vessels, ST Segment.

We need to find how many rows are there, because we don't want a dataset which is too large because it can use so much resources and too small dataset can easily overfit. By drawing some basic plots on raw data, we can find the patterns. It also helps to find that whether the given data is balanced or need to be balanced.

2. Data Cleaning

2.1 Handling Null Values

Medical data is usually very messy and incomplete and most of the rows have at least one or more than one blank value in it. So, data cleaning taken more than 60% of total time. Because we can't just fill the all value with mean, median, mode, we need to choose according to the need of each column. We have other options also such as to choose any complex model to fill missing value but it will become very complex. So, we just choose to fill missing value with 3M value which is Mean, Median, Mode, if any row contains most of its value as null than it is best to drop that row.

2.2 Handling Outliers

In most of the cases we think outlier is any error value so we simply drop that but in medical data we need to work mostly on outlier data because most of the patient have any parameter which is not similar as of any normal person And we are making the model to find the pattern that a person have possibly heart disease or not so we need to consider the outliers also. But we can't consider those outliers which is next to impossible like BP is 10,000, age is 1000 and so on. Because they are next to impossible and impacting the mean value so much. So, we need to handle outliers by removing them or managing its value.

3. Data Preprocessing

3.1 Converting Categorical Value into Numeric Value

ML models can give regression and categorical value as output for which we need to give it a clean dataset which contain only numeric value for training by which it can understand the trends and predict the output for test dataset.

We need to convert string or any other data type into numeric datatype. We can't just assign one value as 1 other as 0 because For more than two categorical values if we constantly increase value than it can affect the output feature because model start to take those value's magnitude also. So, we had usen One-Hot encoding in which converted numeric value don't contain any magnitude with itself.

3.2 Feature Scaling

Scaling is the most crucial part of preprocessing. Because model like KNN and SVM uses the distance formula but if uses the value directly for training dataset than it can use only those feature data which have large value like For Weight (in kg) and Height (in cm) Naturally height is around >150cm and weight lies mostly <100kg so it emphasizes height instead of weight just on the basis of their value So we need to scale them in such a way that training dataset can't get influence by single features so much.

3.3 Consistent Pipeline

One of the key points we need to handle that we need to ensure while training the dataset via different model to get the most appropriate and optimal model that, we need to pass the same preprocessed dataset to each model otherwise it will be a unfair comparison across all models to get the one.

4. Train-Test Splitting

After performing all EDA(Preprocessing) we need to do train-test-split. Where we keep large chunk as training dataset and small chunk as test dataset.

Why we need test dataset?

We want to see how model predict the value for that scenario whose data, it has not seen already. We firstly train the model than check it accuracy for the test data whether it is predicting correct value or not.

This condition mostly arises when model start to memorize the things instead of learning and started to give high performance on training data and low score on testing data. This condition also known as model overfitting.

5. Model Selection and Training

We had implemented five models on same preprocessed training dataset which is used mostly in categorical dataset.

These were Logistic Regression, Naive Bayes, Support Vector Machine, K-Nearest Neighbors and Decision Tree.

We firstly implement that model on our training dataset one by one to check if any problem occurs during training the dataset. And than send the model for Evaluation or checking the score of Model

6. Model Evaluation

To check that our model is how good we just can't rely on accuracy because it can give wrong result for unbalance dataset. For example, if we have 90 true and 10 false and our model learn to give true only still its accuracy is 90% which is very good but this model is unable to give False in any case which is very big problem in real world data. So, here recall count and F1-score need to be counted

7. Score Comparison and Best Model Selection

After getting all Evaluation scores we compare different model where we get the evidence that KNN performs well among all. So, we use KNN model for further deployment and prediction and make more upgradation which can help to improvise KNN model in better way and increase its accuracy

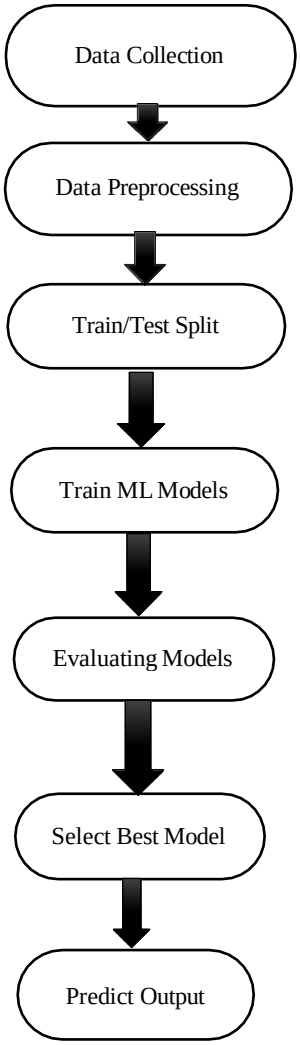


Fig 3.1: Flow of process

IV. RESULTS

After training and evaluating the all model we find that which evaluation parameter is needed to emphasize because each model gives different value for different parameter. Now we need to create a priority list for all Parameters. We considered priority of recall value very much greater than the accuracy because we can't bare the mistake of not detecting the actual potential risk.

Recall > F1 Score > precision > accuracy > False negatives

Here's the table with all the results:

Model	Accuracy	Precision	Recall	F1 score	False Negatives
Logistic Regression	0.81	0.80	0.93	0.87	2
Support Vector Machine	0.85	0.81	0.89	0.85	3
KNN	0.87	0.79	0.96	0.87	1
Naïve Bayes	0.83	0.79	0.96	0.87	2
Decision Tree	0.82	0.76	0.89	0.82	3

Recall Value is use to find in all TRUE value how many models have detected. In our case it becomes very compulsory to not miss any potential heart disease patient detection. Here in the table, we can find Recall Score of KNN and Naive Bayes is highest (0.96) than Logistic Regression Model with (0.93) score and least is Decision Tree and SVM have Score (0.89). So, we don't take Least score model for further comparison.

F1-score is the combined score of Recall and Precision and we can change their priority ratio according to our need Like in our case we can see we are emphasizing more on recall value. In the table we have same F1-score of 0.87 for all three Model (KNN, Naïve Bayes, Logistic Regression).

Precision value tells that out of all True how many are actually true and how many of them are false Positive. It tells out of high risk detected patient how many of them actually have any kind of heart disease. In the table KNN and Naïve Bayes have value (0.79) and Logistic Regression have slightly more value as (0.80). So, we keep model for further evaluation.

Accuracy is the detection of how many predicted values is correct. In the evaluation table we can find that KNN have highest accuracy of (0.87) than Naïve Bayes with the score of (0.83) and at last there come the score of Logistic Regression with the score of (0.81). So, we choose KNN model for further Steps like integration of model and Deployment of model on web server.

For getting the recall and precision value we need the confusion matrix by which we can find the True Positive, True Negative, False Positive, False Negative.

True Positive: Predicted and Actual value is Positive.
True Negative: Predicted and Actual value is Negative.
False Positive: Predicted Positive and Actual value is Negative.
False Negative: Predicted Negative and Actual value is Positive.

The confusion matrix of the best-performing model (KNN) is shown in Figure below

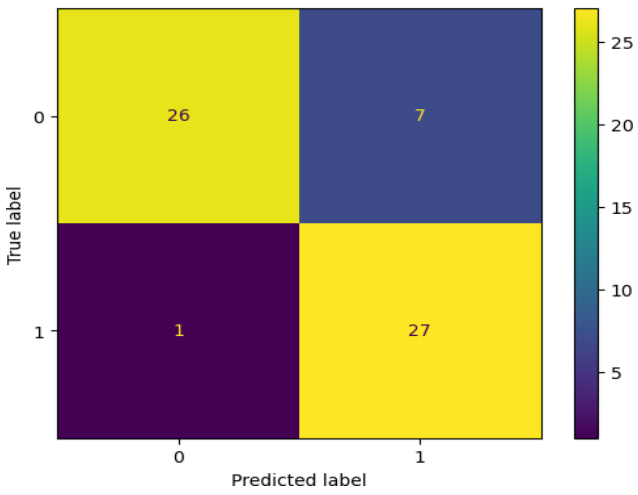


Fig 4.1

A correlation heatmap was drawn to see how features had relationships with the each other. This is shown in Figure

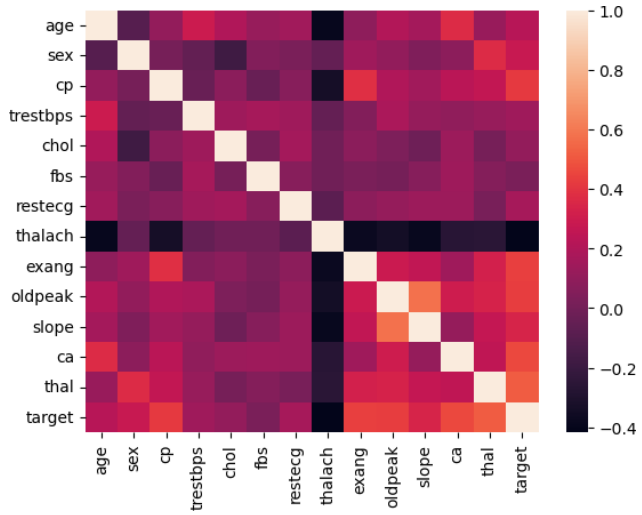


Fig 4.2

V. DISCUSSION

When we look all model in wholesome manner, than we can see that every model has accuracy almost equal which gives us first thought that we can choose any of the trained model. But when we dive deeper, we can find that in our project the most important thing is to

I. find the actual potential risk patient and alert them about their disease and advices them to consult to the doctor than doctor take further action according to requirement.

II. In this whole process to detecting the potential risk patient and tell them to consult doctor if any person doesn't have any disease still detected as high risk than he/she consults to doctor and know that he doesn't need to worry because actually he/she has low risk of heart disease.

In case I and II it is condition I have more priority than II because worst case of our model is to tell them that they are OK but they have any initial stage disease.

So, KNN and Naïve Bayes have very high recall value which shows that they are very good in detecting the actual heart disease patient. Other model is also good but they missed the more positive value than KNN and Naïve Bayes.

Another important point that we find that SVM have more precision value is more than KNN still we not choose SVM because It is important to find risk patient than finding out of total how many are actual correct positive. Because it's ok that we find any wrong positive than to not detect the actual patient.

By applying all metrices evaluation and our case study we can find KNN is the most accurate because it requires not too much fine-tuning. When we combine all the result according to their priority we can find KNN is stronger than Naïve Bayes.

Another Most important insight is we can't just rely on accuracy while building our model we need to consider all other aspect to get the best result/Model.

VI. CONCLUSION

After going though whole process like data cleaning, preprocessing, basic visualization, splitting data into train and test, training the model and evaluate the different parameter score. We can conclude that every model has done their job but we have special requirement that we can compromise with the accuracy but Model need to have high recall value because recall value tells that out of all positive how many positive, we have detected. And in disease detection Unable to detect any potential patient is the last thing that we choose to happen because that patient can be cured at initial stage more easily as compared to later stage.

That most important thing that we realized is that accuracy alone can't matter a lot there are many other methods which work in Model as evaluation parameters help to make it more better model or to select best model among many models.

In our project KNN and Naïve Bayes giving best result in all Parameters but KNN make very less tuning which mean it can give best result in every setting.

Our project showed that using multiple evaluation matrices helps us to get the best model and ML model can help us in early heart disease prediction which help to keep ourselves safe and also consult to doctor for advice.

VII. LIMITATIONS

1. The model is trained on a relatively small public dataset, which may limit its real-world accuracy.
2. The system predicts only the risk of heart disease and cannot replace professional medical diagnosis.
3. Some important medical and lifestyle factors were not included in the dataset.
4. The model performance depends heavily on data preprocessing and feature selection.
5. Real-time hospital data was not used for testing and validation.

VIII. Future Scope

1. The system can be trained on larger and more diverse real-world medical datasets.
2. Advanced deep learning techniques can be implemented to improve prediction accuracy.
3. Wearable devices and IoT-based health monitoring can be integrated into the system.
4. Explainable AI techniques can be added to make predictions easier for doctors to understand.
5. The application can be expanded to support prediction of multiple diseases.

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